PALLAB DEY

Data Analyst | SQL, Excel, Power BI, Python

Sylhet, Bangladesh (Open to relocation & remote) | +880-1751164660 | pallab@pallabdey.com | LinkedIn | GitHub | Portfolio

SUMMARY

Computer Science graduate focused on data analytics, with hands-on project experience uncovering an 11.3% gross margin decline despite revenue growth, identifying 20% of 8,287 e-commerce listings as potentially miscategorized, and developing rule-based risk indicators that flagged a potentially suspicious transaction chain valued at approximately $518K in synthetic mobile-money data. Skilled in SQL, Python, Power BI, and Databricks, with experience in exploratory analysis, data quality assessment, KPI reporting, and dashboard development.

SKILLS

Programming & Analysis: SQL (MS SQL Server, PostgreSQL), Python (Pandas), ETL Pipelines, Data Modeling

BI & Visualization: Power BI, DAX, Excel, Power Query, Dashboard Design

Cloud & Automation: Databricks, n8n, GitHub Actions, Git/GitHub, Vector Databases (Pinecone)

PROJECTS

Maven Toys Sales & Inventory Analysis | SQL Server, Power BI, DAX, Power Query | Live Dashboard | GitHub

- Built an end-to-end sales and inventory analysis on **829K transaction records** across **50 retail stores** by modeling a star-schema in SQL Server, uncovering an **11.3%** YTD decline in gross margin despite **30.9%** revenue growth.
- Designed a **5-page interactive Power BI dashboard** with drill-through store detail pages and custom DAX measures (YoY growth, margin %, stockout risk), giving stakeholders store-level and category-level visibility.
- Identified that the top revenue category (**Toys**, **35%** of revenue) carried the lowest margin (**21%**) while Electronics led on margin, translating the finding into a category-mix and inventory recommendation.

Fintech Transaction Monitoring & Risk Analytics | Databricks (PySpark/SQL), n8n, Telegram Bot API | GitHub

- Engineered a cloud data pipeline in Databricks to process **6.3M synthetic mobile money transactions**, building a star-schema model to support daily and weekly KPI reporting.
- Developed **4 explainable, rule-based risk triggers** (fraud rate breach, transfer-to-cash-out chains, volume collapse, transaction spikes) validated against ground-truth fraud labels, surfacing a high-risk transfer-to-cash-out chain worth **~$518K** (synthetic data).
- Automated reporting with 3 cron-scheduled n8n workflows (daily KPI monitor, weekly summary, conditional alert engine that stays silent unless a risk trigger fires), delivering reports to Telegram with no manual dashboard checks.

Daraz Price & Market Intelligence Pipeline | Python, PostgreSQL (Neon), GitHub Actions, Power BI | Live Dashboard | GitHub

- Built an ETL pipeline loading **8,287** scraped Daraz listings into Neon PostgreSQL as 7 simulated daily batches, with GitHub Actions running 4 times a day to process each batch and log rows cleaned, flagged and dropped per run.
- Designed a TF-IDF similarity check that **flagged 1,650 mis-filed listings (~20%)** using per-category thresholds and a reviewed override list.
- Analyzed **8,252 cleaned listings across 15 categories** in a published 2-page Power BI dashboard; excluding 1,650 flagged listings raised the **median TV price from 17,200 to 32,250 Taka (+87.5%).**

EDUCATION

Metropolitan University, Sylhet | 2021 - 2025

B.Sc. in Computer Science and Engineering — CGPA: 3.96 / 4.00

CERTIFICATIONS

Data Analyst Associate | DataCamp

Data Analyst in Databricks | DataCamp

ACHIEVEMENTS

- Recipient of the Chairman's Scholarship and Vice Chancellor's Scholarship at Metropolitan University.
- Solved **100+** SQL problems across StrataScratch, DataLemur, HackerRank, and LeetCode — documented in the SQL Practice Portfolio on GitHub.